

Dendrology

Science

May, 2026



**SE
TU**

Ollscoil
Teicneolaíochta
an Oirdheiscirt

South East
Technological
University

Short Title: Dendrology
Faculty Science and Computing
Department Science
Cluster Forestry
Author MSOTTOCORNOLA
Credits 5

Level: Introductory

Description of Module / Aims

This module includes the identification, classification, nomenclature, ecology and geographic range of the major tree species of Ireland, and the more common understory vegetation associated with forestry. Students will learn to use dichotomous keys in the field identification of trees. This module will also provide a comprehensive knowledge of the silviculture and uses of the main Irish forest tree species.

Programmes

FORS-0002 BSc in Forestry (WD_SFORS_D)

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Indicative Content

- Introduction to dendrology
- Rules and use of scientific classification and nomenclature
- Building and use of dichotomous plant keys
- Conifers and broadleaves (three field trips)

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify the major tree species of Ireland from living specimens as well as from samples of flowers, twigs, leaves, fruits, and written descriptions.
2. Use rules of scientific nomenclature to correctly present the common and binomial names.
3. Employ dichotomous plant keys to identify unknown species.
4. State major and minor uses of each species.
5. Describe physical, biological, ecological and silvicultural features associated with the major tree/forest types in Ireland.

Learning and Teaching Methods

- Lectures.
- Demonstrations.
- Assigned reading.
- Hands-on classes.
- Field study.
- Preparation of a 20 species herbarium.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
<i>In-Class Assessment</i>	60%	1,2,4,5
<i>Project</i>	20%	1,2,3
<i>Assignment</i>	20%	1,2,3

Assessment Criteria

- <40%:** Very limited knowledge and understanding of the subject matter. Lacking in technical ability/skill and understanding to carry out assessments. Failure to meet the objectives of assessments. Unable to carry out or submit independent study and research. Unable to present written work with attention to correct formatting, grammar and spelling.
- 40%-49%:** Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on assessments. Attempt made to meet objectives and carry out independent study and research. Final result lacks balance and arguments undeveloped. Can present written work correctly formatted with minor grammar and spelling errors only.
- 50%-59%:** Demonstrate satisfactory general knowledge of the main issues within the subject. Logically and competently carry out assessments. Satisfactory handling and awareness in carrying out independent study and research. Reporting and projects adequately structured and referenced.
- 60%-69%:** Demonstrate sound knowledge of the subject matter. Show ability to analyse and logically argue in an effective and mature style. Demonstrate initiative and the ability to criticise methods used and appreciation of experimental design and current knowledge limitations when carrying out assessments. Projects and reporting coherent and soundly structured illustrating wide and in depth reading on subject coupled with the proper use of references.
- 70%-100%:** Demonstrate authoritative handling of complex material and a highly developed and extensive understanding of the subject. Provide well focused analysis and convincing argument on the subject. Demonstrate imaginative approach, excellent technical ability, awareness, and aptitude in carrying out and reporting on assessments. Show initiative, individual thought, and enthusiasm in self-study and independent learning.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the theory and practical components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Field Trips	24	
Independent Learning	87	

Essential Material

- "Forest Service Irish forest Species Information Sheets:Department of Agriculture, Food and the Marine." <http://www.agriculture.gov.ie/forestservice/publications>

- Horgan, T., M. Keane, R. Mc Carthy, M. Lally and D. Thompson. *_A guide to forest Tree species__Selection and Silviculture in Ireland_*. Ireland: COFORD, 2003.
- Mitchell, A. *_A field guide to the trees of Britain and Northern Europe_*. UK: Collins, 1974.
- Parnell, J.A.N., T. Curtis and E.L. Cullen. *_Webb's An Irish flora_*. Ireland: Cork University Press, 2012.

Supplementary Material

- Hardin, J., D. Leopold and F.M. White. *_Textbook of Dendrology_*. 9th ed. USA: Mc Graw Hill, 2001.
- Hogan, S. *_Trees for all seasons_*. UK: Timber press, 2008.
- Joyce, P.M. *_Growing Broadleaves_*. Ireland: COFORD, 1998.
- Savill, P.S. *_The Silviculture of trees used in British forestry_*. 2nd. UK: CABI, 2013.
- Tyler , M. and B. Farley. *_British oaks - a concise guide_*. UK: Crowood Press, 2008.
- Vaucher, H. *_Tree Bark: A colour guide_*. UK: Timber Press, 2003.